

Intégration:

Primitive et intégrale d'une fonction continue

Def: soit f et F fonctions telles que:
 f soit continue sur $I \subset \mathbb{R}$
 F soit dérivable sur I

On dit que F est une primitive de f sur I

$$\text{Si: } \forall x \in I: F'(x) = f(x)$$

Remarques:

1] Si F est une primitive de f sur I alors les fonctions $F + c$ sont des primitives de f sur I (avec $c \in \mathbb{R}$)

2] On note la primitive F de f par:

$$F(x) = \int f(x) dx$$

Proposition:

Si f est continue sur $[a, b]$ et F est une primitive de f sur $[a, b]$ alors:

$$\int_a^b f(x) dx = [F(x)]_a^b = F(b) - F(a)$$

Propriétés:

$$1] \int_a^a f(x) dx = 0$$

$$2] \int_a^b f(x) dx = - \int_b^a f(x) dx$$

$$3] \int_a^b f'(x) dx = [f(x)]_a^b = f(b) - f(a)$$

1) si $c \in]a; b[$ alors : $\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$

2) si f_a est paire ; alors

$$\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$$

3) si f est impaire ; alors : $\int_{-a}^a f(x) dx = 0$

4) $\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx$

5) si : $f(x) \leq g(x)$ alors : $\int_a^b f(x) dx \leq \int_a^b g(x) dx$

Exemple :

soit $f(x) = (2x-1)e^{-x}$

Déterminer a, b :

pour que : $x \mapsto (ax+b)e^{-x}$ est une primitive
est une primitive de f

Proposition : soit $a \in \mathbb{R}$ et $u = u(x)$ une
fonction.

$$a \longrightarrow ax + C$$

$$x^n \longrightarrow \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$$

$$\frac{1}{x} \longrightarrow \ln|x| + C$$

$$\frac{1}{\sqrt{x}} \longrightarrow 2\sqrt{x} + C$$

avec $C \in \mathbb{R}$

$\frac{u'}{u} \rightarrow \ln u + C$	}	$u' \cos u \rightarrow \sin u + C$
$\frac{u'}{\sqrt{u}} \rightarrow 2\sqrt{u} + C$		$u' \sin u \rightarrow -\cos u + C$
$\frac{u'}{u^n} \rightarrow \frac{-1}{(n-1)u^{n-1}} + C$		$\frac{u'}{1+u^2} \rightarrow \arctan u + C$
$u' \cdot u^n \rightarrow \frac{u^{n+1}}{n+1} + C$		$\frac{u'}{\sqrt{1-u^2}} \rightarrow \arcsin u + C$
$u' e^u \rightarrow e^u + C$		$\frac{-u'}{\sqrt{1-u^2}} \rightarrow \arccos u + C$

Exemple: calculer les intégrales suivantes :

1] $\int (1 - 2x + \frac{2}{1-x}) dx$

2] $\int [\cos(1-3x) - 2x(3x^2-1)^3 + \frac{7}{(x-1)^3}] dx$

$u' \cos u \rightarrow \sin u$

$\cos(1-3x) \rightarrow -\frac{1}{3} (-3 \cos(1-3x))$

$\cos(1-3x) \rightarrow -\frac{1}{3} (\sin(1-3x))$

$u' (u^3) \rightarrow \frac{u^4}{4} \Leftrightarrow -2x(3x^2-1)^3$

$-2x(3x^2-1)^3 = \frac{1}{3} (3(2x))(3x^2-1)^3$ avec $u = 3x^2-1$
 $u' = 6x$

$= \frac{1}{3} (6x)(3x^2-1)^3 = \frac{1}{3} \frac{(3x^2-1)^4}{4}$

$\frac{u'}{u^2} \rightarrow \frac{-1}{2(u^2)}$ avec $u = x-1$
 $u' = 1$

$\frac{7}{(x-1)^3} = 7 \left(\frac{1}{(x-1)^3} \right) \rightarrow 7 \left(\frac{-1}{2(x-1)^2} \right)$

$$3) K = \int_0^1 e^x \sqrt{e^x - 1} dx$$

$$u' \cdot u^n = \frac{u^{n+1}}{n+1} \quad \text{donc} \quad u' u^{\frac{1}{2}} = \frac{u^{\frac{1}{2}+1}}{\frac{1}{2}+1} = \frac{u^{\frac{3}{2}}}{\frac{3}{2}}$$

$$u' u^{\frac{1}{2}} \rightarrow \frac{2}{3} u^{\frac{3}{2}} \quad ; \quad e^x \sqrt{e^x - 1} \rightarrow \frac{2}{3} (e^x - 1)^{\frac{3}{2}}$$

avec $\begin{cases} u = e^x - 1 \\ u' = e^x \end{cases}$

$$K = F(1) - F(0) = \frac{2}{3} (e - 1)^{\frac{3}{2}}$$

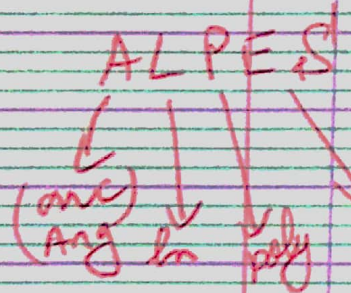
2' intégration par partie:

soient $u = u(x)$ et $v = v(x)$ deux fonctions

alors: $\int u(x) v'(x) dx = u(x)v(x) - \int u'(x)v(x) dx$

et $\int_a^b u v' dx = [u \cdot v]_a^b - \int_a^b u' v dx$

Exemple: 1) $\int (x-1) e^x dx$



SinX
COSX
parfois : tanx

avec $\begin{cases} u(x) = x-1 \\ v(x) = e^x \end{cases} \Rightarrow \begin{cases} u'(x) = 1 \\ v(x) = e^x \end{cases}$

$$I = (x-1) e^x - \int e^x dx = (x-1) e^x - e^x + C$$

$$I = (x-2) e^x + C$$

$$\text{Ex } I = \int \ln(1+x^2) dx = \int \underbrace{x}_{u'} \ln(1+x^2) dx$$

$$\text{I.P.P: } \int u(x) v'(x) dx = u(x) v(x) - \int u'(x) v(x) dx$$

$$\text{avec } \begin{cases} u(x) = \ln(1+x^2) \\ v'(x) = 2x \end{cases} \Rightarrow \begin{cases} u'(x) = \frac{2x}{1+x^2} \\ v(x) = x^2 \end{cases}$$

$$\text{Donc } I = x^2 \ln(1+x^2) - \int \frac{2x^2}{1+x^2} dx$$

$$= x^2 \ln(1+x^2) - 2 \int \frac{x^2}{1+x^2} dx$$

$$\begin{array}{r} x^2 \overline{) x^2+1} \\ -x^2-1 \\ \hline 2 \end{array}$$

$$\Rightarrow \frac{x^2}{x^2+1} = 1 - \frac{1}{x^2+1}$$

$$\begin{array}{c} P \\ \hline R \end{array} \bigg| \begin{array}{c} Q \\ \hline R \end{array}; \quad d^0 R < d^0 Q$$

$$\text{donc: } \frac{P}{Q} = 1 + \frac{R}{Q}$$

calculons:

$$I_1 = \int \frac{x^2}{1+x^2} dx = \int \left(1 - \frac{1}{x^2+1} \right) dx$$

$$1 \rightarrow x$$

$$\frac{1}{x^2+1} \rightarrow -(\arctan x)$$

$$\begin{cases} \frac{u'}{u^2+1} \rightarrow \arctan u \\ \frac{1}{x^2+1} \rightarrow \arctan x \end{cases}$$

$$\text{donc } I_1 = x - \arctan x$$

$$I = x^2 \ln(1+x^2) - 2(x - \arctan x)$$

$$K = \int e^{2x} \cos(3x) dx$$

$$\text{I.P.P.} \quad \int u(x) \cdot v'(x) dx = u(x)v(x) - \int u'(x)v(x) dx$$

$$\text{avec } \begin{cases} u(x) = e^{2x} \\ v'(x) = \frac{1}{3} (3 \cos 3x) \end{cases} \Rightarrow \begin{cases} u'(x) = 2e^{2x} \\ v(x) = \frac{1}{3} (\sin 3x) \end{cases}$$

$$u' \cos u \rightarrow \sin u$$

$$\text{avec: } \begin{cases} u = 3x \\ u' = 3 \end{cases}$$

$$\text{donc: } K = \frac{1}{3} e^{2x} \sin 3x - \frac{2}{3} \int e^{2x} \sin 3x dx$$

On applique I.P.P. $\sin K_1$

$$\begin{cases} u(x) = e^{2x} \\ v'(x) = (3 \sin 3x) \frac{1}{3} \end{cases} \Rightarrow \begin{cases} u'(x) = 2e^{2x} \\ v(x) = \frac{1}{3} (-\cos 3x) \end{cases}$$

$$\text{donc } K_1 = -\frac{1}{3} e^{2x} \cos 3x + \frac{2}{3} \int e^{2x} \cos 3x dx$$

$$K_1 = -\frac{1}{3} e^{2x} \cos 3x - \frac{2}{3} K$$

$$K = \frac{1}{3} e^{2x} \sin 3x - \frac{2}{3} \left(-\frac{1}{3} e^{2x} \cos 3x + \frac{2}{3} K \right)$$

$$\Rightarrow K = \frac{1}{3} e^{2x} \sin 3x + \frac{2}{9} e^{2x} \cos 3x - \frac{4}{9} K$$

Intégrale de la forme: $I = \int \frac{1}{ax^2+bx+c} dx$

On a: $ax^2+bx+c = a \left[\left(x + \frac{b}{2a}\right)^2 - \frac{\Delta}{4a} \right]$

On remplace: (On pose: $t = x + \frac{b}{2a}$)

1) $\int \frac{1}{t^2+R^2} dx = \frac{1}{R} \arctan\left(\frac{t}{R}\right) + C$

2) $\int \frac{1}{t^2} dx = -\frac{1}{t} + C$

3) $\int \frac{1}{t^2-R^2} dx = \frac{1}{2R} \ln \left| \frac{t-R}{t+R} \right| + C$

Exemple:

1) $\int \frac{1}{x^2+x+1} dx$

$\begin{cases} a=1 \\ b=1 \\ c=1 \\ \Delta=-3 \end{cases}$

On a: $x^2+x+1 = \left(x + \frac{1}{2}\right)^2 + \frac{3}{4}$

On remplace:

$\int \frac{1}{\left(x + \frac{1}{2}\right)^2 + \frac{3}{4}} dx = \frac{1}{R} \arctan\left(\frac{t}{R}\right) + C$
 $= \frac{2}{\sqrt{3}} \arctan\left(\frac{2}{\sqrt{3}} \left(x + \frac{1}{2}\right)\right) + C$

2) $\int \frac{2}{x^2-6x+9} dx = 2 \int \frac{1}{x^2-6x+9} dx$ $\begin{cases} a=1 \\ b=-6 \\ c=9 \\ \Delta=0 \end{cases}$

On a:

$x^2-6x+9 = (x-3)^2$

On remplace:

$\int \frac{2}{(x-3)^2} dx = -\frac{2}{(x-3)} + C$

3) $k = \int \frac{1}{3x^2+6x+2} dx$ $\begin{cases} a=3 \\ b=6 \\ c=2 \\ \Delta=12 \end{cases}$

On a: $3x^2+6x+2 = 3 \left[(x+1)^2 - \frac{1}{3} \right]$

On remplace:

$k = \frac{1}{3} \int \frac{1}{(x+1)^2 - \frac{1}{3}} dx = \frac{1}{3} \int \frac{1}{(x+1)^2 - \left(\frac{1}{\sqrt{3}}\right)^2} dx$

avec $\begin{cases} t = x+1 \\ R = \frac{1}{\sqrt{3}} \end{cases}$ $= \frac{1}{3} \frac{1}{2R} \ln \left| \frac{t-R}{t+R} \right| + C$
 $= \frac{\sqrt{3}}{6} \ln \left| \frac{x+1 - \frac{1}{\sqrt{3}}}{x+1 + \frac{1}{\sqrt{3}}} \right| + C$

Intégrale de la forme I: $\int \frac{P(x)}{ax^2+bx+c} dx$

$$1) \int \frac{u'}{u} dx = \ln |u|$$

$$2) \int \frac{1}{ax^2+bx+c} dx$$

Exemple: $I = \int \frac{2x-3}{x^2+x+2} dx$

On pose: $u = x^2+x+2$
 $u' = 2x+1$

$$I = \int \frac{2x+1-4}{x^2+x+2} dx$$

$$I = \int \frac{2x+1}{x^2+x+2} dx - 4 \int \frac{1}{x^2+x+2} dx$$

$$I = \ln |x^2+x+2| - 4 I_1$$

calculons $I_1: \int \frac{1}{x^2+x+2} dx$

On a:

$$x^2+x+2 = \left(x+\frac{1}{2}\right)^2 + \frac{7}{4} = \left(x+\frac{1}{2}\right)^2 + \left(\frac{\sqrt{7}}{2}\right)^2$$

$$I_1 = \int \frac{1}{t^2+R^2} dx = \frac{1}{R} \arctan\left(\frac{1}{R}\right)$$

$$I_1 = \frac{2}{\sqrt{7}} \arctan\left(\frac{x+\frac{1}{2}}{\frac{\sqrt{7}}{2}}\right) = \frac{2}{\sqrt{7}} \arctan\left(\frac{2x+1}{\sqrt{7}}\right)$$

On remplace dans I:

$$I = \ln(x^2+x+2) - \frac{8}{\sqrt{7}} \arctan\left(\frac{2x+1}{\sqrt{7}}\right) + C$$

Intégrale de la forme $\int \frac{1}{\sqrt{ax^2+bx+c}} dx$

On a: $ax^2+bx+c = a \left[\left(x+\frac{b}{2a}\right)^2 - \frac{\Delta}{4a^2}\right]$

On remplace dans I:

$$1) \int \frac{1}{\sqrt{t^2+k^2}} dx = \operatorname{arsh}\left(\frac{t}{k}\right) + C$$

$$2) \int \frac{1}{\sqrt{k^2-t^2}} dx = \operatorname{arcsin}\left(\frac{t}{k}\right) + C$$

$$3) \int \frac{1}{\sqrt{t^2-k^2}} dx = \operatorname{arcsh}\left(\frac{t}{k}\right) + C$$

$$I = \int \frac{-2}{\sqrt{x^2 - 3x + 4}} dx$$

$$\begin{cases} a=1 \\ b=-3 \\ c=4 \\ \Delta=-7 \end{cases}$$

$$x^2 - 3x + 4 = \left(x - \frac{3}{2}\right)^2 + \frac{7}{4}$$

$$= \left(x - \frac{3}{2}\right)^2 + \left(\frac{\sqrt{7}}{2}\right)^2$$

On remplace:

$$I = -2 \int \frac{1}{\sqrt{t^2 + R^2}} = -2 \operatorname{arcsinh} \left(\frac{t}{R} \right)$$

$$I = -2 \operatorname{arcsinh} \left(\frac{2x-3}{\sqrt{7}} \right) + C$$

$$2 \int \frac{1}{\sqrt{1-x^2+2x+1}} dx$$

$$\text{On a: } -x^2 + 2x + 1 = -\left[(x-1)^2 - 2\right]$$

$$= 2 - (x-1)^2$$

$$= (\sqrt{2})^2 - (x-1)^2 \quad ; \quad K = \sqrt{2}$$

$$t = x - 1$$

On remplace:

$$J = \int \frac{1}{\sqrt{K^2 - t^2}} dx = \operatorname{arcsin} \left(\frac{t}{K} \right)$$

$$= \operatorname{arcsin} \left(\frac{x-1}{\sqrt{2}} \right) + C$$

Intégrale de la forme $\int \frac{f(x)+q}{\sqrt{ax^2+bx+c}} dx$

$$1) \int \frac{u'}{\sqrt{u}} dx = 2\sqrt{u}$$

Exemple:-

$$K = \int \frac{3x-7}{\sqrt{2x^2 - \frac{4}{3}x - \frac{2}{3}}} dx$$

$$\text{On a: } 2x^2 - \frac{4}{3}x - \frac{2}{3} = a \left[\left(x - \frac{b}{2a}\right)^2 - \frac{\Delta}{4a^2} \right]$$

$$= 2 \left[\left(x - \frac{1}{3}\right)^2 - \frac{4}{9} \right]$$

$$\begin{cases} a=2 \\ b=-\frac{4}{3} \\ c=-\frac{2}{3} \\ \Delta=\frac{64}{9} \end{cases}$$

$$\text{On pose } u = 2x^2 - \frac{4}{3}x - \frac{2}{3}$$

$$u' = 4x - \frac{4}{3}$$

$$K = \frac{3}{4} \int \frac{\frac{4}{3}(3x-7)}{\sqrt{2x^2 - \frac{4}{3}x - \frac{2}{3}}} dx$$

$$K = \frac{3}{4} \int \frac{4x - \frac{28}{3}}{\sqrt{2x^2 - \frac{4}{3}x - \frac{2}{3}}} dx$$

$$K = \frac{3}{4} \int \frac{(4x - \frac{41}{3}) - \frac{24}{3}}{\sqrt{2x^2 - \frac{4}{3}x - \frac{2}{3}}} dx$$

$$K = \frac{3}{4} \left[\int \frac{4 - \frac{4}{3}}{\sqrt{2x^2 - \frac{4}{3}x - \frac{2}{3}}} dx - \frac{24}{3} \int \frac{1}{\sqrt{2x^2 - \frac{4}{3}x - \frac{2}{3}}} dx \right]$$

$$= \frac{3}{4} \left[2 \sqrt{2x^2 - \frac{4}{3}x - \frac{2}{3}} - \frac{24}{3} K_1 \right]$$

Intégrale de la forme $\int f(e^x) dx$:

On pose $t = e^x \Rightarrow dt = e^x dx$

$$\Rightarrow dt = t dx \Rightarrow dx = \frac{1}{t} dt$$

Exemple:

$$I = \int_0^1 \frac{1}{e^x + 3e^{-x}} dx \quad \text{on pose: } t = e^x$$

$$dx = \frac{1}{t} dt$$

$$\Rightarrow I = \int_1^e \frac{1}{t + 3t^{-1}} \frac{1}{t} dt = \int_1^e \frac{1}{t + \frac{3}{t}} \frac{1}{t} dt$$

$$\text{Pose: } x=0 \Rightarrow t = e^0 = 1$$

$$x=1 \Rightarrow t = e^1 = e$$

$$I = \int_1^e \frac{1}{t^2 + 3} dt = \int_1^e \frac{1}{t^2 + (\sqrt{3})^2} dt$$

$$\int \frac{1}{t^2 + R^2} = \frac{1}{R} \arctan\left(\frac{t}{R}\right)$$

$$= \left[\frac{1}{\sqrt{3}} \arctan\left(\frac{t}{\sqrt{3}}\right) \right]_1^e = \frac{1}{\sqrt{3}} \left[\arctan\left(\frac{e}{\sqrt{3}}\right) - \arctan\left(\frac{1}{\sqrt{3}}\right) \right]$$

Intégrale de la forme: $\int f(\cos x) \sin x dx$

On pose: $t = \cos x \Rightarrow dt = -\sin x dx$
 $\Rightarrow -dt = \sin x dx$

Int de la forme: $\int f(\sin x) \cos x dx$

On pose: $t = \sin x \Rightarrow dt = \cos x dx$

Int de la forme: $\int f(\sin x, \cos x) dx$

On pose: $t = \tan\left(\frac{x}{2}\right)$; $\cos x = \frac{1-t^2}{1+t^2}$
 $\sin x = \frac{2t}{1+t^2}$; $dx = \frac{2}{1+t^2} dt$

exemple:

1) $I = \int \frac{\sin^3 x}{2 + \cos x} dx$
 $\sin^3 x = \sin^2 x \sin x$
 $\sin^2 x = (1 - \cos^2 x)$

$I = \int \frac{1 - \cos^2 x}{2 + \cos x} \sin x dx$

On pose: $t = \cos x \Rightarrow dt = -\sin x dx$
 $\sin x dx = -dt$

$I = \int \frac{1-t^2}{2+t} (-dt) = \left(\frac{t^2-1}{t+2} \right) dt$

$$\begin{array}{r|l} t^2-1 & t+2 \\ -t^2-2t & \\ \hline -2t-1 & \\ 2t+4 & \\ \hline 3 & \end{array}$$

$\frac{P}{R} \Big/ \frac{Q}{S} \Rightarrow \frac{P}{Q} = S + \frac{R}{Q}$

Donc: $\frac{t^2-1}{t+2} = t-2 + \frac{3}{t+2}$

$I = \int \left[t-2 + 3 \left(\frac{1}{t+2} \right) \right] dt = \frac{t^2}{2} - 2t + 3 \ln|t+2| + C$

$I = \frac{\cos^2 x}{2} - 2\cos x + 3 \ln|\cos(x)+2| + C$

• type

example Derive Int to form $\int f(\sin x, \cos x) dx$

$$I = \int \frac{1}{1 - \sin x} dx$$

On pose $t = \tan\left(\frac{x}{2}\right)$

$$\sin x = \frac{2t}{1+t^2}$$

$$dx = \frac{2}{1+t^2} dt$$

$$I = \int \frac{1}{1 - \frac{2t}{1+t^2}} \cdot \frac{2}{1+t^2} dt = \int \frac{2}{t^2 - 2t + 1}$$

$$I = 2 \int \frac{1}{t^2 - 2t + 1} dt = 2 \int \frac{1}{(t-1)^2} dt$$

$$\int \frac{1}{u^2} = \frac{-1}{u} \quad , \quad \frac{d}{dt}(t-1) = 1$$

$$I = 2 \left(\frac{-1}{t-1} \right) = \frac{-2}{t-1} = \frac{-2}{\tan\left(\frac{x}{2}\right) - 1} + C$$